

Adjunctive Therapies with Topical Calcineurin Inhibitors

Adjunctive therapies such as emollients should be continued. When applied after bathing, pimecrolimus or tacrolimus should be applied first to affected areas, followed by emollients to unaffected skin. Application site stinging from topical calcineurin inhibitors (TCIs) may be more pronounced in areas with excoriation. Pre-treating these areas with a topical corticosteroid for 2 to 7 days may reduce irritation and improve adherence to therapy.

Imiquimod

Imiquimod should be applied sparingly to affected areas when treating warts or superficial basal cell carcinoma (BCC). For actinic keratosis on the face or scalp, a broader but well-defined treatment area is recommended—limited to one region at a time. For anogenital warts, initial application should be three times per week. If there is little or no response after 2 to 4 weeks, increasing to daily use may improve efficacy, though this also increases the risk of irritant dermatitis. Application should occur before bedtime and the medication should remain on the skin for 6 to 10 hours. Patients must be informed that imiquimod does not eliminate the risk of human papillomavirus transmission to physical contacts. The frequency of application varies by indication: five times weekly for superficial BCC and two times weekly for actinic keratosis.

Monitoring Therapy

Pimecrolimus and Tacrolimus

If effective, topical calcineurin inhibitors (TCIs) can be incorporated into the long-term management of atopic dermatitis. Physicians should ensure that patients use these agents intermittently and only for active dermatitis. Adverse events, particularly application site reactions, should be monitored. Patients should be closely evaluated for cutaneous infections, with special attention to eczema herpeticum, which occurred at a slightly increased rate in pre-marketing trials. Any lymphadenopathy should be documented and followed closely. Regular skin examinations are prudent to monitor for signs of cutaneous malignancy. No routine serologic monitoring is required; however, if systemic immunosuppression is suspected, serum levels of tacrolimus or pimecrolimus may be measured. If blood levels are obtained, residual medication—especially from common application sites like the antecubital fossa—must be thoroughly removed prior to venipuncture to avoid falsely elevated results. Despite reassuring safety data, the long-term safety of TCIs has not been definitively established. Ongoing vigilance for potential adverse effects, including immunosuppression and malignancy, is warranted.

Imiquimod

For warts, imiquimod therapy should continue until complete clearance or up to 4 months before considering treatment failure. If there is no response or clinical worsening at 8 weeks, alternative therapies should be considered. With partial response, continuation of therapy may still be beneficial. If well tolerated, frequency may be increased from three times weekly to daily. No serologic monitoring is needed. Irritant reactions can be managed with temporary discontinuation and, if necessary, topical anti-inflammatory treatment. Therapy can then be restarted at a reduced frequency. Similar principles apply for superficial BCC and actinic keratosis, though these conditions require closer clinical follow-up.